



nopCommerce + OpenTelemetry

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The Problem

Flow: The customer places an order.

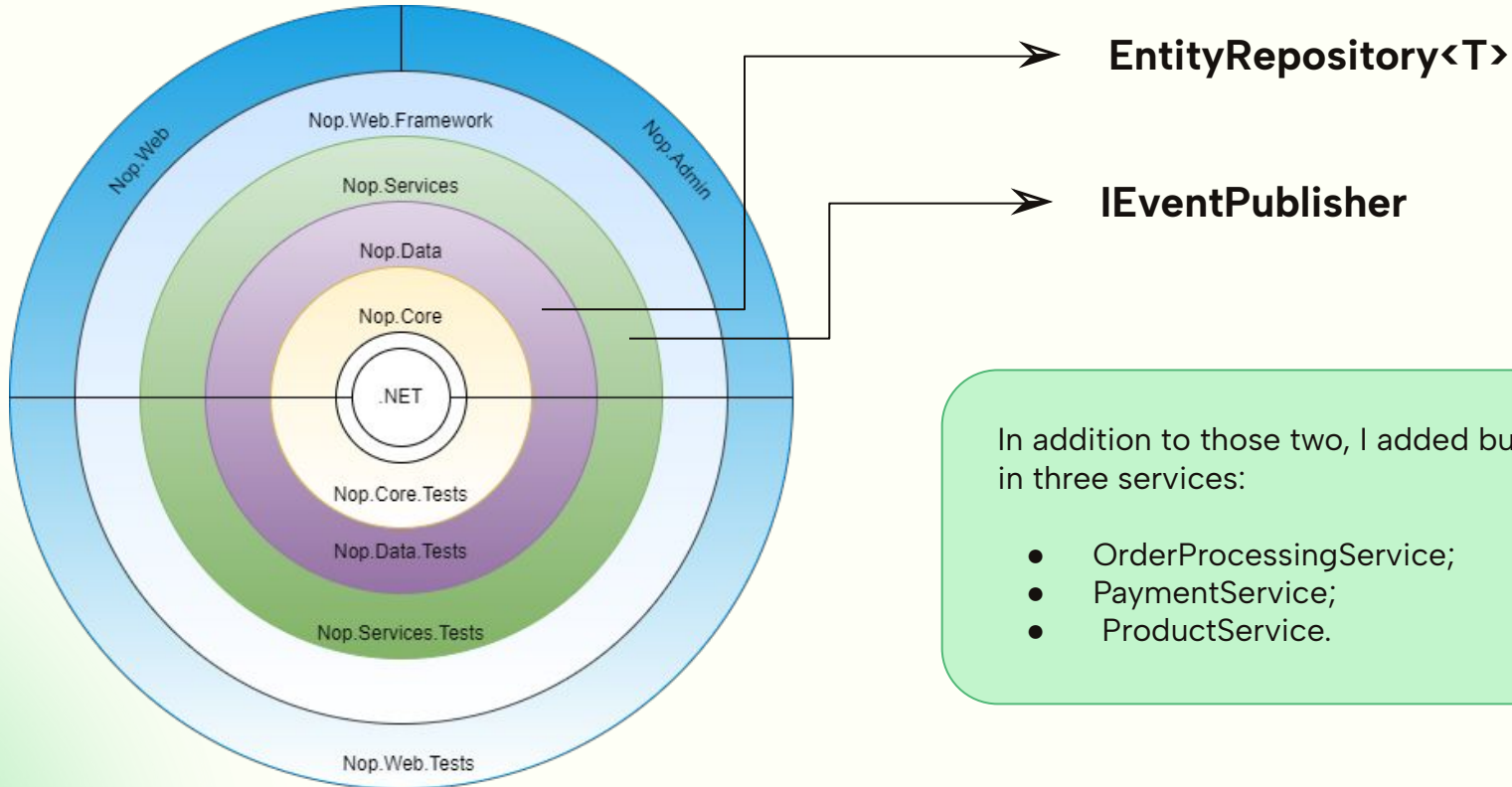
No Observability

- **Slow checkout:** Nobody knows why;
- **Payment fails silently:** Detectable by mail;
- **Out of stock:** Only detected if you go to see it.
- **p95 rises during peak:** cause? database? cache? plugin?
Unknown

With observability

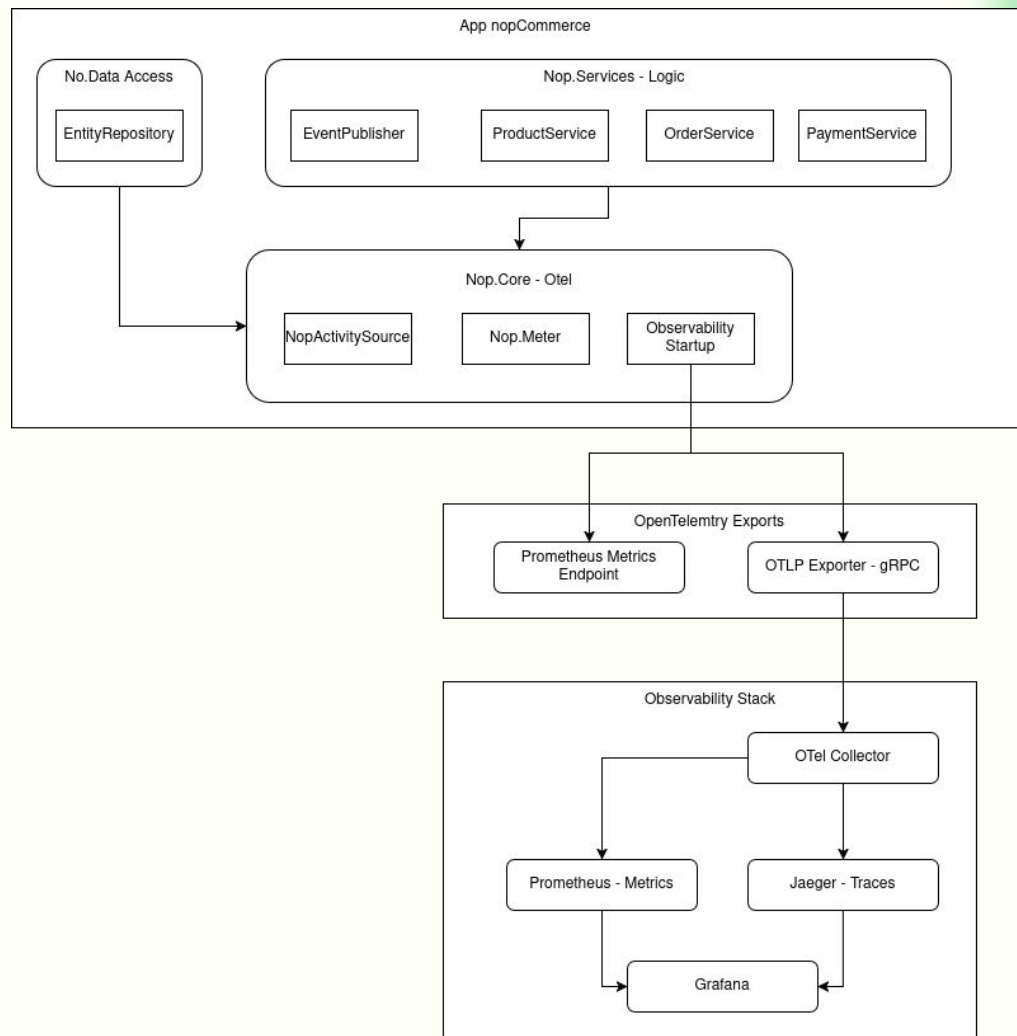
- **Full trace:** HTTP → Service → Database → Events;
- **Payment failure** visible in <10 seconds;
- **Stockout** detected in **real time**.
- **Latency** attributed to the **exact layer** that caused it.

Architecture and instrumentation points



In addition to those two, I added business spans in three services:

- OrderProcessingService;
- PaymentService;
- ProductService.



Spans and Metrics

Spans

nop.order.place: Groups payment, database, and events as children.

nop.payment.process isolates the external component.

nop.inventory.adjust: Registers stock_after by product.

db.insert / update / delete: One span per write operation

The event <EventName> records that the event has been published.

Metrics

Nop.basket.checkout_started: Detects cart abandonment before checkout is finalized.

nop.order.placed Primary KPI: Success/Failure Rate by Payment Method

nop.checkout.duration SLO: checkout p95 < 3s

nop.payment.result: Silent payment failures

nop.payment.duration isolates external gateway latency.

...

Privacy Strategy (PII)

PII appears in browsing, cart, checkout, and payment

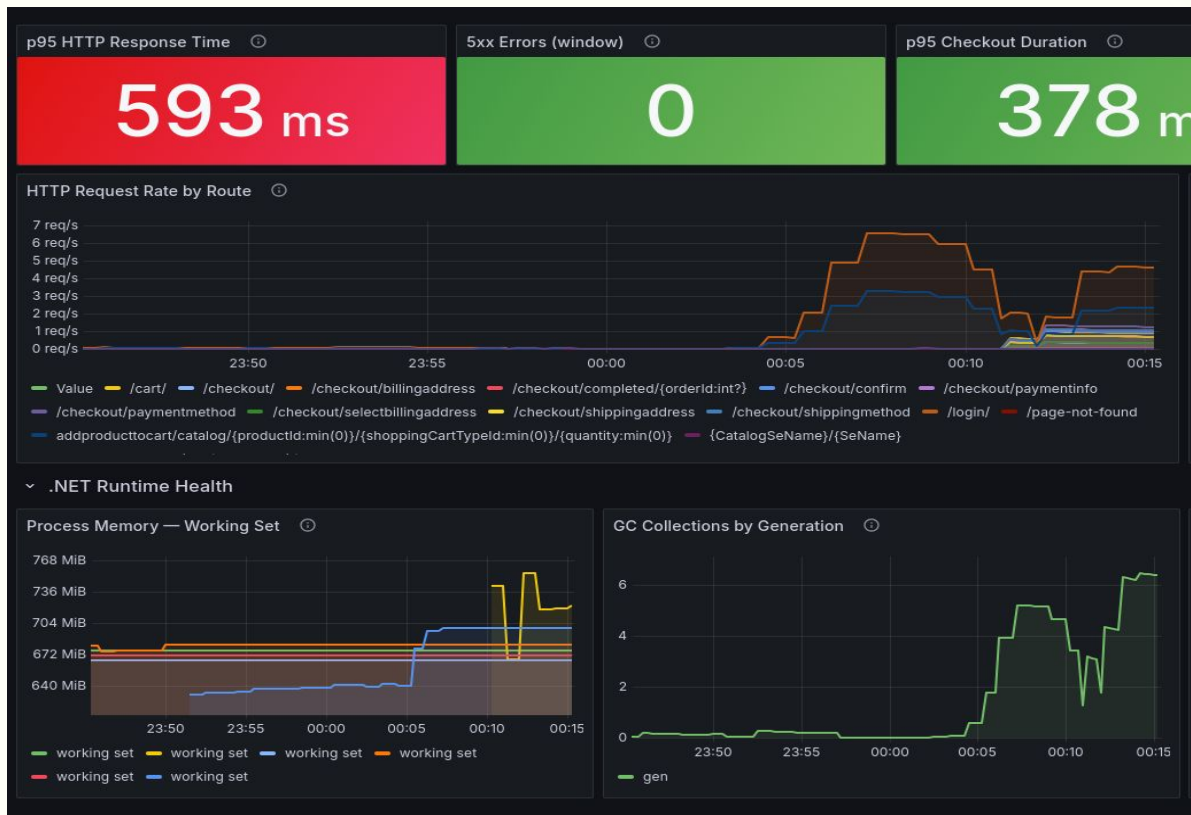
Layer 1 — Code

- No PII attribute is added to spans.:
 - Email;
 - Address;
 - Phone Number;
 - Order total.

Layer 2 — OTel Collector

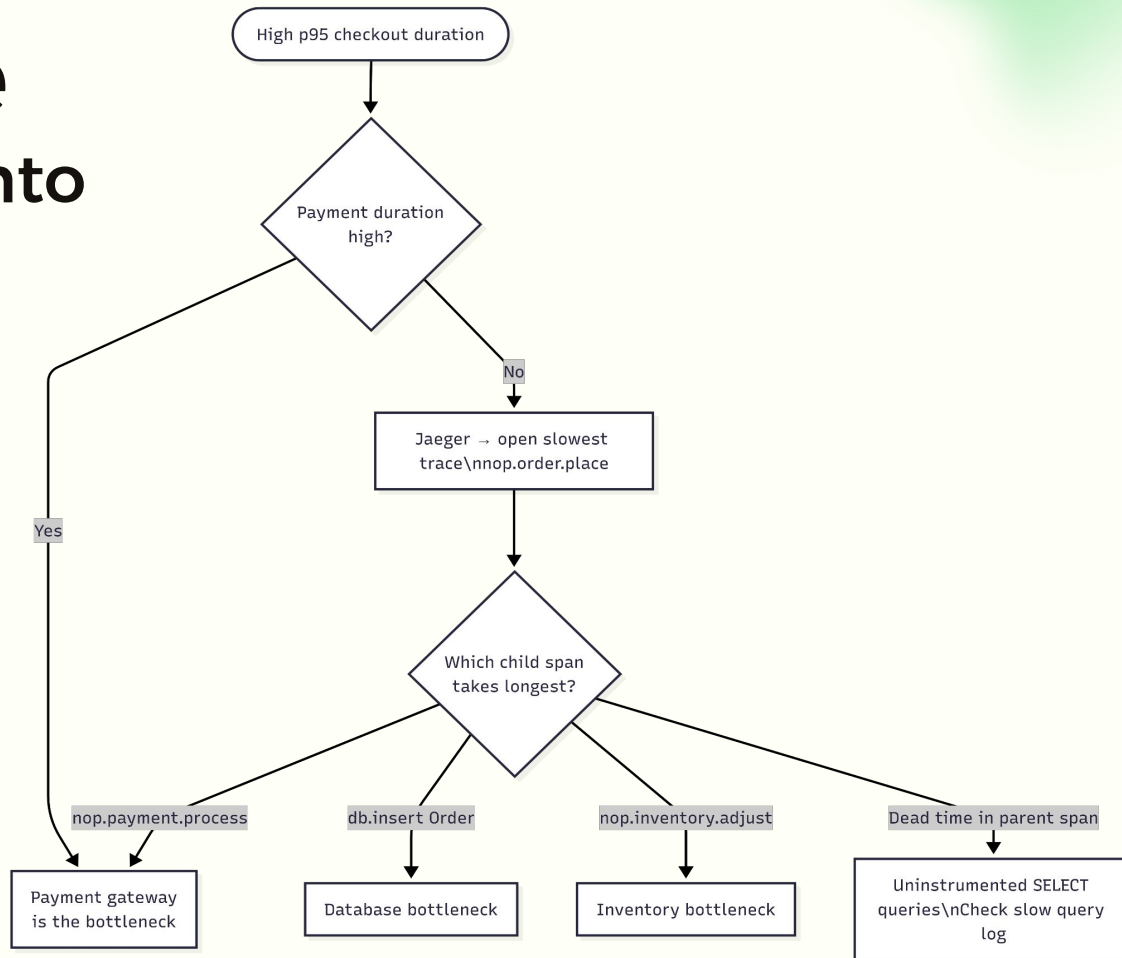
- Processor transform/ `sanitize_pii` in the Collector pipeline;
- Remove sensitive attributes before exporting to Jaeger/Prometheus.

Use Case - Checkout lento

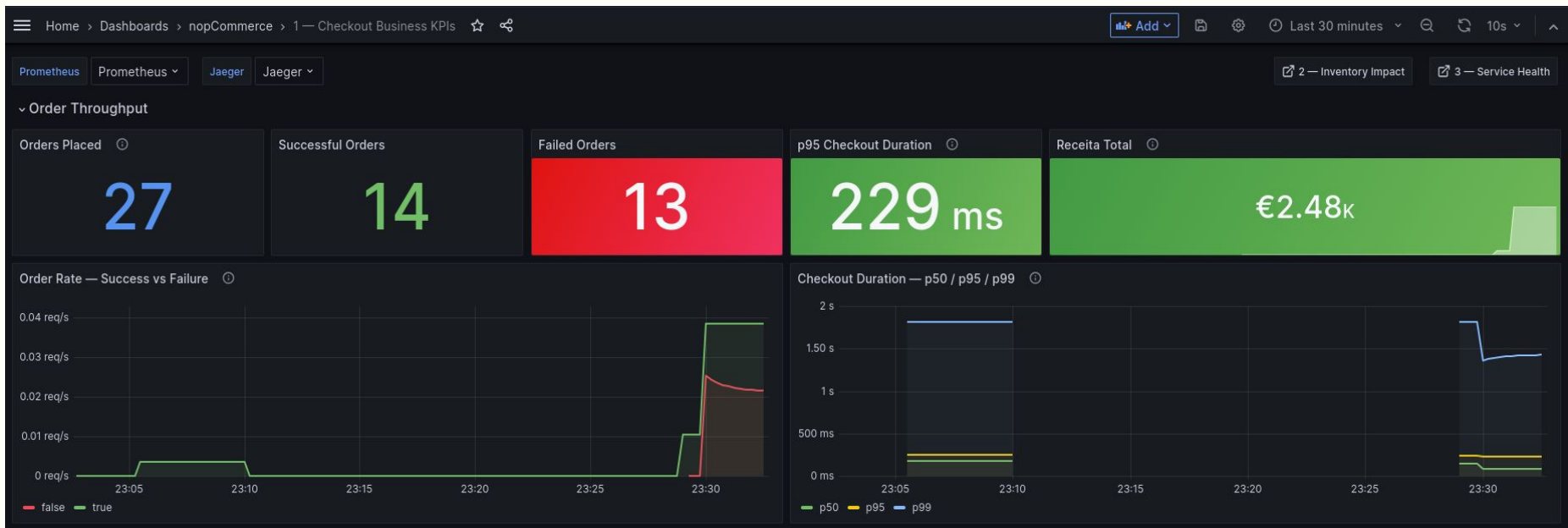


Use Case

Checkout lento

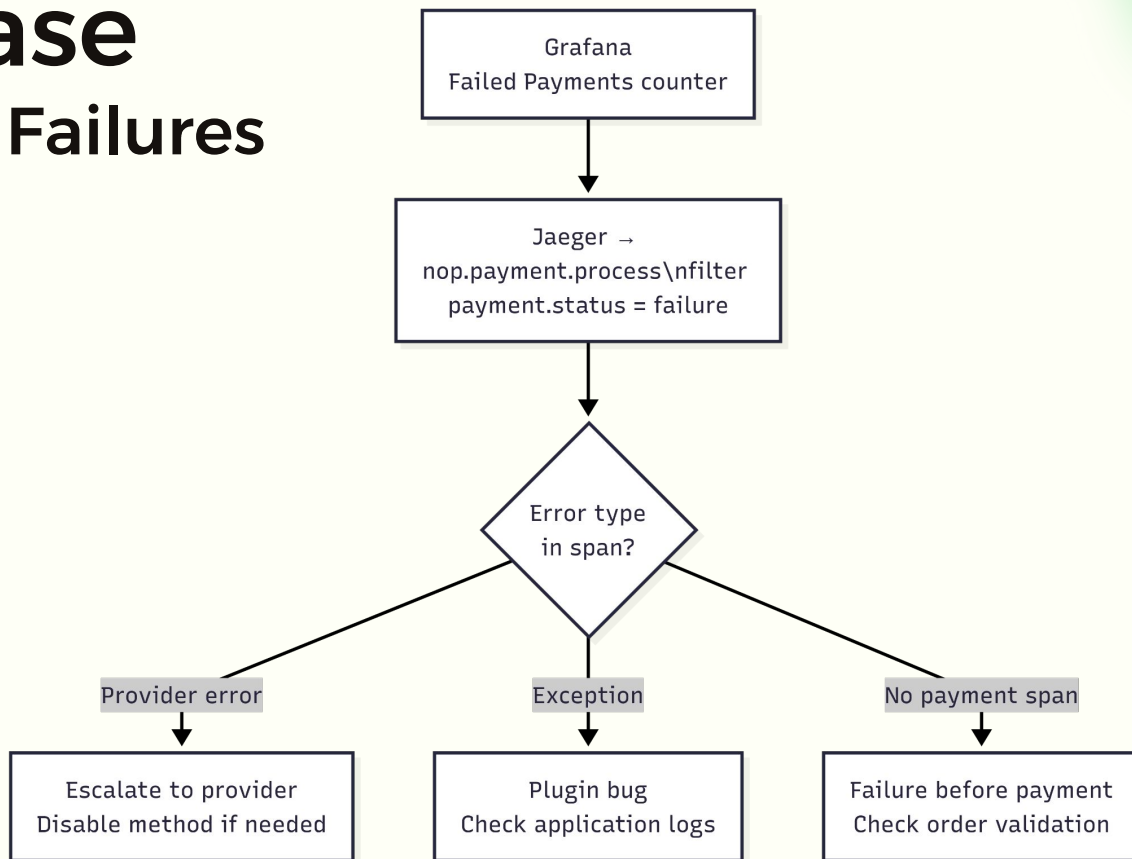


Use Case - Payment Failures

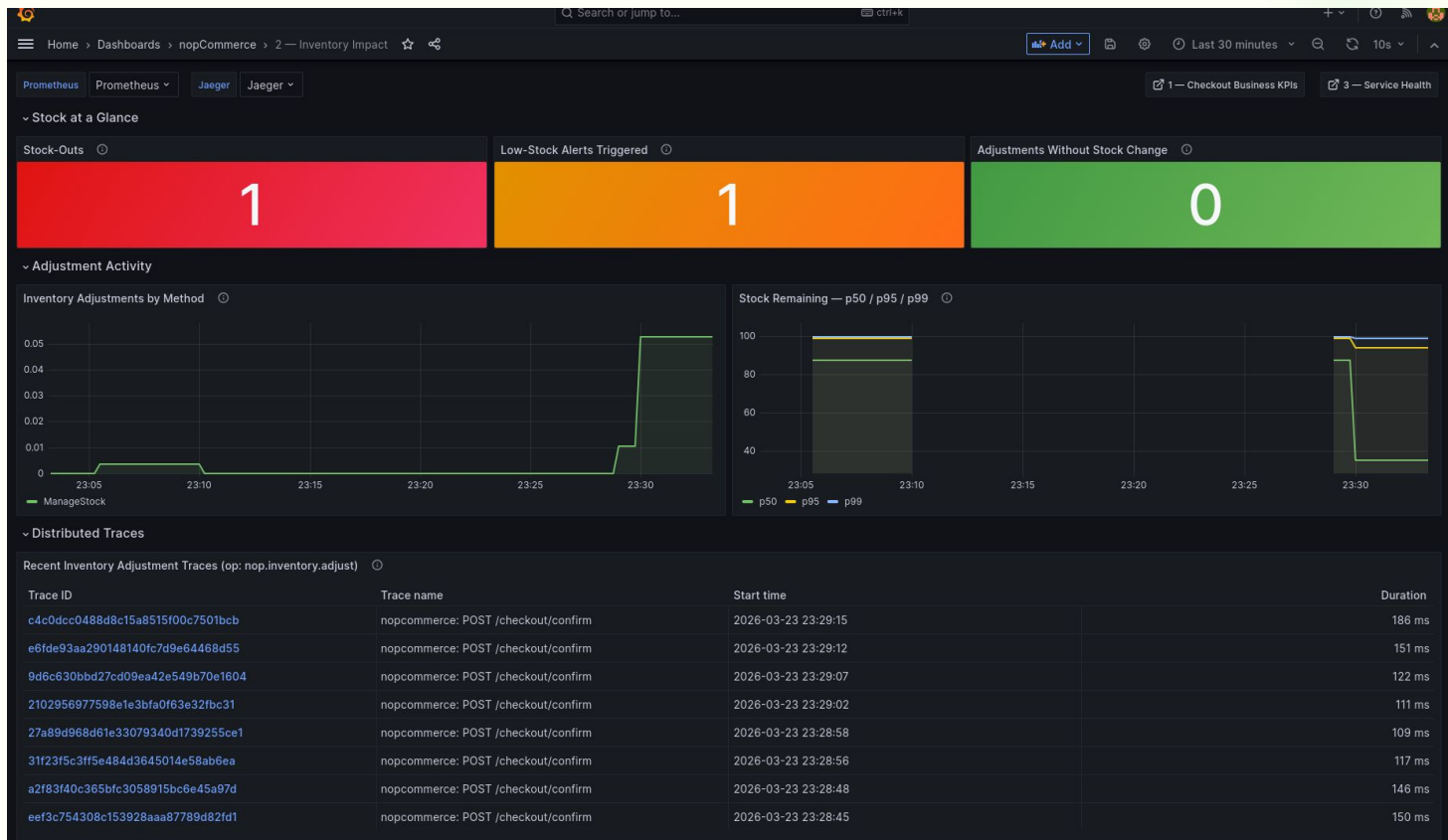


Use Case

Payment Failures



Use Case - Out of stock



TRADEOFFS

SELECT operations

The method that prepares the data before creating a **request executes dozens of SELECT** queries. Instrumenting each call would require **modifying hundreds of methods**.

Individual event handlers

The class retrieves its dependencies from a central register itself, instead of receiving them injected from the outside.

Plugins (Payment system)

External assemblies loaded at runtime do not generate spans.